Ex. No. 3 — Designing a Quarter-Wave Transformer by Varying a) Frequency b) Input Impedance c) Characteristic Impedance

Test Case 1 — Varying Frequency

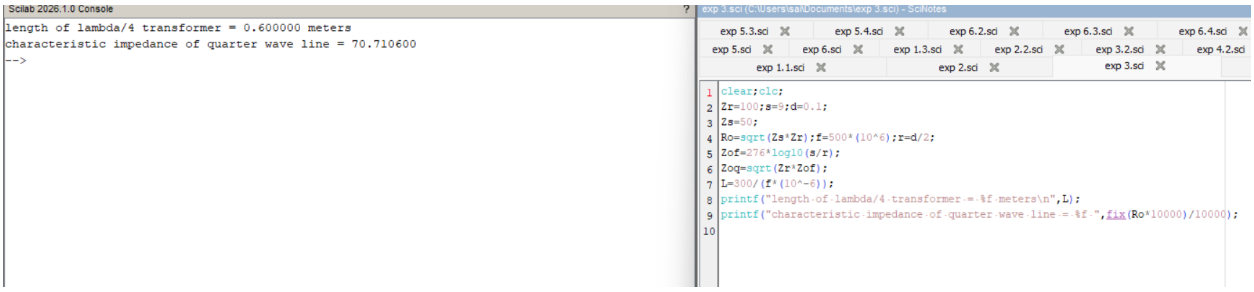
Design a quarter-wave transformer for an input impedance Z_{in} = 50 Ω and a load impedance Z_L = 100 Ω. Analyze the transformer's performance as the operating frequency is varied across {100 MHz, 200 MHz, 300 MHz, 500 MHz}.

Scilab Program

```
clear; clc;
Zr = 100; Zs = 50;
f = 100*10^6;              // set frequency per row (Hz)
Ro = sqrt(Zs*Zr);
L = 300/(f*10^-6);
printf('length of lambda/4 transformer = %f meters\n', L);
printf('characteristic impedance of quarter wave line = %f ', fix(Ro*10000)/10000);
```

Sample Console Output

```
length of lambda/4 transformer = 0.600000 meters
characteristic impedance of quarter wave line = 70.710600
-->
```



Characteristic Impedance of the Quarter-Wave Line

Z■ (ohms)	Z■ (ohms)	Simulation R■ (ohms)	Theoretical R■ (ohms)
50	100	70.7107	70.7107

Length of λ/4 Transformer vs Frequency (Z_L = 100 Ω, Z_{in} = 50 Ω)

Frequency (MHz)	Simulation Length (m)	Theoretical Length (m)
100	3.0000	3.0000
200	1.5000	1.5000
300	1.0000	1.0000
500	0.6000	0.6000

Test Case 2 — Varying Input Impedance

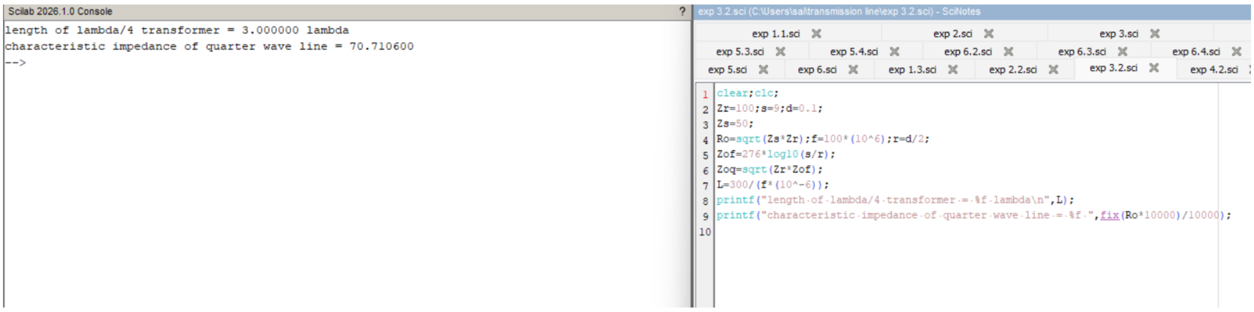
For a fixed operating frequency of 2 GHz and a load impedance Z_L = 100 Ω, design a quarter-wave transformer and evaluate its performance as the input impedance Z_{in} varies through {25 Ω, 50 Ω, 75 Ω, 100 Ω}. Validate that the transformer's characteristic impedance ($Z_{transformer} = R_o = \sqrt{(Z_{in} \cdot Z_L)}$) changes with Z_{in}.

Scilab Program

```
clear; clc;
Zr = 100; f = 2*10^9;
Zs = 25;                    // set Zin per row
Ro = sqrt(Zs*Zr);
L = 300/(f*10^-6);
printf('length of lambda/4 transformer = %f lambda\n', L);
printf('characteristic impedance of quarter wave line = %f ', fix(Ro*10000)/10000);
```

Sample Console Output

```
length of lambda/4 transformer = 3.000000 lambda
characteristic impedance of quarter wave line = 70.710600
-->
```



R■ vs Z_{in} (Z_L = 100 Ω, f = 2 GHz)

Z■■ (ohms)	Simulation R■ (ohms)	Theoretical R■ (ohms)
25	50.0000	50.0000
50	70.7107	70.7107
75	86.6025	86.6025
100	100.0000	100.0000

Result

A quarter-wave transformer is designed at a fixed operating frequency of 2 GHz and load impedance (Z_L) of 100 Ω, and its performance is evaluated for various input impedance values. The variation of the transformer's characteristic impedance with Z_{in} is also validated.